Jorge "George" Chumbipuma

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Education

- **Rice University** — Houston, TX
  Doctor of Philosophy – Computational and Applied Mathematics — *August 2023 – May 2028 (Expected)*
  GPA: *3.95/4.0*
  Courses: *Advanced Numerical Analysis, Modeling Mathematical Physics, Applied Functional Analysis, Numerical Methods for PDEs, High-Performance Computing, Systems of Equations & Unconstrained Optimization*

- **San José State University** — San Jose, CA
  Master of Science – Mathematics — *August 2020 – August 2022*
  GPA: *3.9/4.0;* ***Honors:*** *Phi Kappa Phi*
  Courses: *Advanced Matrix Theory, Numerical Partial Differential Equations, Stochastic Processes, Numerical Linear Algebra*

- **University of California, Irvine** — Irvine, CA
  Bachelor of Science – Double Major in Electrical Engineering and Physics — *September 2010 – June 2013*
  Minor: *Information and Computer Science;* ***Honors:*** *Tau Beta Pi*
  Courses: *Data Structures, Numerical Analysis, Statistical Physics, Engineering Probability, Machine Vision, Computer Organization*

Certificates

- **De Anza College** — Cupertino, CA
  Certificate of Achievement – Programming in Python — *June 2020 – August 2022*

Skills

- **Programming Languages**: Python, Java, C/C++, Julia, MATLAB, R (Scientific Computing), SQL, LaTeX, Bash Scripting
- **Libraries/Frameworks**: NumPy, Pandas, Matplotlib, BeautifulSoup, SQLite, Axelrod, MASS, Git, Makefile, OrdinaryDiffEq.jl
- **Tools & Platforms**: Mathematica, Jupyter, SSH, GitHub, Windows, MacOS, Linux (Ubuntu), Raspberry Pi, Rice NOTS Cluster, SLURM
- **Software Development**: Memory Management, Matrix Operations, Sparse Matrix Representation, OOP, Data Structures, Algorithms, Parallel Programming (CUDA, OpenMP, MPI)
- **Mathematical & Computational Skills**: Computational Mathematics, Numerical Methods for PDEs, Optimization Techniques, Scientific Computing in Julia, Finite Difference Methods, Roofline Model Analysis

Experience

- **AoPS (Art of Problem Solving) Academy** — Remote (United States)
  Virtual Math Instructor (Part-time) — *September 2023 – Present*
  - **Educate** and **inspire** high-achieving K-12 students through AoPS's unique problem-solving curriculum in a virtual setting.
  - **Facilitate** interactive, participation-oriented classes focusing on skill development rather than rote memorization.
  - **Manage** small class sizes (average of 12 students) of top-performing, passionate learners.

- **Pumatics** — Remote (United States)
  Founder & Lead Educator — *January 2022 – Present*
  - **Designed** and **implemented** a comprehensive curriculum that caters to diverse learning needs, ensuring consistent engagement across all tutoring sessions.
  - **Fostered** strategic collaborations with educators, parents, and consultants, achieving a remarkable 90% student satisfaction rate.

- **MIT Lincoln Laboratory** — Boston, MA
  Summer Research Intern (Full-time) — *May 2024 – August 2024*
  - **Conduct** research on the modeling and analysis of Intelligence, Surveillance, and Reconnaissance (ISR) and tactical systems.
  - **Develop** mathematical models of system performance based on physical principles and engineering.
  - **Implement** models in MATLAB to perform simulations and analyses.
  - **Collaborate** with a multidisciplinary team to address mission-relevant questions posed by senior leaders in the Department of Defense.
  - **Present** research findings and progress to senior researchers and stakeholders, contributing to critical decision-making processes.

- **Doerr Institute for New Leaders** — Rice University, Houston, TX
  Professional Leadership Coaching (Part-time) — *August 2023 – December 2023*
  - Engaged in a semester-long, one-on-one coaching program with an International Coaching Federation (ICF) certified coach, focusing on personal leadership development.

- Received and implemented constructive feedback to enhance leadership skills, focusing on areas like empathic listening, effective delegation, and trust-building.
- Improved assertiveness and reduced impulsiveness through targeted behavioral practices, significantly strengthening emotional intelligence domains.

- **Elite Educational Institute** San Jose, CA
  Instructor (Part-time) *February 2020 – July 2023*
  - **Instructed** multiple students in a small group tutoring or classroom setting (class sizes ranged from 5 to 25 students).
  - **Designed** math and physics classes, provided class material, handled attendance, and other academic record-keeping.
  - **Received** 95% positive feedback from students for competence, approachability, and conveying ideas effectively.
- **San José State University** San Jose, CA
  Lecturer (Part-time) *August 2022 – December 2022*
  - **Implement** numerous assignments, exams, and other assessment tools to accurately evaluate student learning and progress for a minimum of 30 students per semester.
  - **Collaborate** with six other faculty members and staff to develop and implement multiple curricular and programmatic initiatives.
  - **Provide** regular academic support and guidance to students, including at least 2 hours of scheduled office hours per week and individual consultations as needed for a minimum of 5 students per semester.
- **San José State University** San Jose, CA
  Lead Teaching Associate (Part-time) *August 2021 – May 2022*
  - **Instructed** over 60 college students in online courses from the Department of Mathematics and Statistics, utilizing multiple educational approaches and classroom activities through online platforms such as Zoom and Miro.
  - **Collaborated** with a team of 7 TAs and three professors to design and implement online coursework for differential, integral, and multivariable calculus courses to meet the needs of undergraduate students.
  - **Organized** online class materials using Google Workspace, including documenting solutions to worksheets using Mathematica and LaTeX.

Projects

- **Numerical Solutions to the Nonlinear Vlasov Equations**: Collaborated with another student on solving the nonlinear Vlasov equations using various numerical methods. Developed and implemented first- and second-order upwind schemes, and Heun's method to model plasma dynamics, with discretization of spatial and velocity domains. Simulated both 1D and 2D cases, with special focus on Vlasov-Poisson and Vlasov-Maxwell systems, using finite difference methods. Presented results and visualizations of linear and nonlinear plasma interactions. Tech: MATLAB, Python, LaTeX. (April 2024)
- **GPU-Based Optimization for Reduction, Stencil, and Matrix Multiplication**: Implemented and optimized GPU kernels using CUDA for reduction, stencil, and matrix-matrix multiplication tasks. Developed three versions of the reduction algorithm to improve parallelization, eliminating idle threads and utilizing shared memory for increased performance. Applied roofline model analysis to benchmark operational intensity and GFLOPS, achieving improvements in reduction and stencil performance using shared memory. Optimized five versions of matrix-matrix multiplication, leveraging block tiling and coalesced memory accesses to maximize computational efficiency across large matrices. Conducted performance comparisons using CUDA profiling tools and generated roofline plots. Tech: CUDA, GCC, nvprof, Roofline plots, Julia, Rice NOTS. (April 2024)
- **MPI-Based Parallel Matrix Multiplication**: Developed parallel implementations of SUMMA and Cannon's algorithms for matrix-matrix multiplication using MPI. Utilized `MPI_Comm_split` for row and column communicators in the SUMMA algorithm to enable efficient block-wise matrix multiplication. Implemented point-to-point communication in Cannon's algorithm using `MPI_Sendrecv` to optimize data exchange across processors. Verified the correctness of parallel results by comparing with serial matrix multiplication. Assessed parallel scalability by varying the number of processors and matrix sizes. Conducted performance testing on Rice University's NOTS cluster and analyzed Frobenius norm differences for accuracy. Tech: MPI, Julia, GCC, OpenMPI, SLURM, NOTS. (April 2024)
- **OpenMP Optimization and Parallel Efficiency**: Implemented and optimized parallel algorithms using OpenMP to enhance the performance of matrix-matrix multiplication and back solve operations. Compared `#omp parallel for` with `#omp parallel for collapse(2)` for blocked matrix multiplication and assessed parallel efficiency by increasing thread count. Developed recursive block matrix multiplication using OpenMP task-based parallelism, verified correctness, and improved performance through dynamic scheduling. Tested static and dynamic scheduling for parallel back solve with unit upper triangular matrices and analyzed efficiency across different matrix sizes. Incorporated non-unit upper triangular matrix handling in the algorithm for further generalization. Tech: C++, OpenMP, Julia, NOTS, GCC. (March 2024)
- **Serial Optimization for Matrix-Matrix Multiplication**: Implemented and optimized various matrix-matrix multiplication algorithms using C++ for high-performance computing. Developed a naive matrix-matrix multiplication, cache-blocked algorithm, and recursive matrix multiplication method. Conducted performance testing and generated roofline plots for different matrix sizes on the NOTS cluster. Determined optimal block sizes for cache-blocking and recursion cutoff thresholds for recursive methods. Compared performance across methods using GFLOPS and execution time. Incorporated Strassen's algorithm for comparison with standard approaches. Tech: C++, Julia, LaTeX, NOTS, GCC, Julia, Roofline plots. (March 2024)

- **Sparse Matrix Operations in Julia**: Implemented a custom `SparseMatrixCSR{T} <: AbstractMatrix{T}` matrix format in Julia, mirroring a C++ implementation but with 1-indexing. Specialized key functions like `Base.size`, `Base.getindex`, and `LinearAlgebra.mul!` for efficient matrix operations. Developed an iterative solver function `iterate(A, b, u, a; tol = 1e-3)` for the equation $u += a * r$ where $r = b - A * u$, and conducted verification using Julia's `@code_warntype`. Constructed a tridiagonal matrix to model a discretized Poisson equation under Neumann boundary conditions, and visualized the solution using Julia's plotting libraries. Incorporated performance analysis with the `@time` macro and compared dense versus sparse matrix implementations. Extended tasks for graduate students included solving a system of ODEs using `OrdinaryDiffEq.jl`, comparing different explicit time-stepping methods, and creating animations of the solutions. Ensured reproducibility through `Project.toml` files. Tech: Julia, OrdinaryDiffEq.jl, LinearAlgebra, Plots. (December 2023)
- **Matrix-Vector Multiplication in C++ (Dense and Sparse Matrices)**: Implemented dense row-major and sparse CSR format matrices in C++ as part of learning matrix-vector multiplication. Developed `DenseRowMatrix` and `SparseMatrixCSR` classes as public derived classes of an abstract matrix class. Both classes featured constructors, destructors, and overloaded operators for indexing and matrix-vector multiplication. Implemented the iteration $u += a * r$, where $r = b - A * u$, in both dense and sparse matrix formats, to solve a discretized Poisson equation under Neumann boundary conditions. Utilized chrono for timing analysis and reported iterations, norm of r, and computation time for both dense and sparse matrices. Extended tasks for graduate students included non-allocating iterations and templating for float types. Tech: C++, Makefile, Git. (November 2023)
- **Matrix-Vector Multiplication in C**: Developed a comprehensive program in C for matrix-vector multiplication, focusing on different memory layouts and performance analysis. Defined a `Matrix` struct and implemented functions for initializing and freeing matrices with both contiguous and non-contiguous memory. Wrote functions for matrix-vector and transpose matrix-vector multiplications. Verified correctness through a custom `verification.c` file, checking against known results. Utilized `time.h` for performance timing, analyzing average runtimes for different matrix sizes and memory layouts. Extended the assignment with additional tasks for graduate students, including implementing the `matvec` function in MATLAB, comparing runtimes with C, and analyzing the impact of GCC's `-O3` optimization flag. Results and analysis were documented in a detailed LaTeX report, including a runtime comparison table. Tech: C, MATLAB, Bash, LaTeX, Git. (October 2023)
- **Automated LaTeX Document Generation and Version Control Integration**: Developed a shell script and LaTeX document for automated document generation and version control using Git and SSH. Script automates LaTeX compilation, managing auxiliary files in a hidden directory and demonstrating error handling for user errors. Utilized SSH for remote access to Rice University's Linux cluster. Authored a LaTeX document detailing SSH process, script functionality, and critique, including sections on potential script issues and improvements. Demonstrated Git proficiency in repository management, tagging, and collaboration. Tech: Bash, LaTeX, Git, SSH. (September 2023)
- **Dynamical Systems in General Relativity**: Focused on replicating mathematical structures from the foundational paper "Dynamical systems applied to cosmology: dark energy and modified gravity" by Sebastian Bahamonde et al. (2018) in the context of Einstein's general theory of relativity. Meticulously reproduced mathematical derivations and calculations, addressing minor discrepancies and typographical errors, contributing to research verification. Applied dynamical systems techniques to gain insights into long-term behavior of cosmological models described by differential equations. Tech: Mathematica, LaTeX. (May 2022)
- **EM-algorithm for a Gaussian Mixture Model (GMM)**: Designed and implemented an Expectation-Maximization (EM) algorithm for a Gaussian Mixture Model (GMM) using the R programming language. Utilized the MASS library and Iris flower data set to provide clear documentation and examples of the algorithm's functionality. Tech: R, Mathematica. (December 2021)
- **Machine Learning in Game Tournaments**: Developed a machine learning project focused on game tournaments, implementing and training a finite state machine player using an evolutionary algorithm for an iterated prisoner's dilemma tournament. Utilized the Axelrod framework to analyze and improve the performance of the machine learning model in this competitive environment. Tech: Python, Linux Shell, Git. (November 2021)
- **Simulating the Gravitational Dynamics of Three-Body Systems of Particles**: Simulated the gravitational dynamics of three-body systems of particles using a range of numerical methods and Python programming. Implemented algorithms such as Explicit Euler, Adams-Bashforth, Runge-Kutta, and Midpoint Rule using NumPy and Matplotlib to generate highly accurate solutions to nonlinear ordinary differential equations. Demonstrated expertise in numerical methods and Python programming in this project. Tech: Python. (October 2021)
- **Introduction to Control Theory**: Explored the basics of control theory in mathematics, including its application in automatically controlled systems such as rocket launches, self-driving cars, and more. Presented an overview of the concepts and results of control theory, as well as common applications and an example. Tech: Mathematica, LaTeX. (May 2021)
- **High Accuracy Computations in Numerical Linear Algebra**: Explored numerical linear algebra techniques, including Cholesky decomposition, QR algorithm, and singular value decomposition (SVD), using MATLAB functions. Measured computational time for determining the smallest eigenvalue of a 100×100 Cauchy matrix using determinant and inverse methods. Successfully calculated the smallest eigenvalue of a 200×200 Pascal matrix within a tenth of a second. Tech: MATLAB, Mathematica. (May 2021)
- **Finite Difference Methods in Solving Reynolds Equations**: Analyzed the pressure distribution under the slider of industry data storage readers by modeling the steady-state using the Reynolds equation. Produced graphical visualizations of the pressure distribution by implementing the finite difference method called successive over-relation (SOR) in MATLAB. Tech: MATLAB, Mathematica. (December 2020)
- **Cubic Non-Linear Schrödinger (NLS) Equation**: Conducted a review of the well-posedness of the cubic nonlinear Schrödinger (NLS) equation, providing physical context and an informal derivation of the equation in the context of intense laser beams in an isotropic bulk medium. Discussed various NLS models for nonlinear optics and Bose-Einstein condensates. Tech: Mathematica. (November 2020)

Presentations

- **Numerical Solutions to the Nonlinear Vlasov Equations**: Presented a collaborative research project titled "Numerical Solutions to the Nonlinear Vlasov Equations" with fellow student Gerardo Daniel Villalobos. This project focused on the numerical solution of the Vlasov equations, a fundamental framework in plasma physics and astrophysics, using various discretization techniques and time-stepping schemes. We applied finite difference methods and explored upwind schemes, forward Euler, and Heun's method to solve the Vlasov-Poisson system in one and two dimensions.

  The presentation highlighted the development and implementation of numerical algorithms to model particle distributions in phase space and their interactions via self-consistent fields. We showcased our simulation results, including animations of particle dynamics and convergence analyses, comparing different numerical methods for stability and accuracy.

  The study also contributed insights into nonlinear systems and their numerical approximations in the context of plasma physics. (Rice University, Houston, Texas, May 2024)

- **Master's Project Defense**: In this presentation, I showcased my master's writing project, "Dynamical Systems in General Relativity," undertaken as part of the Project option requirement. Guided by my project advisor, Dr. Slobodan Simić, the project delved into applying dynamic systems theory within the general relativity (GR) framework to comprehend and model cosmological phenomena. This research addressed the critical need for a comprehensive theoretical framework capable of elucidating the observed accelerated expansion of the universe—an epochal discovery made by Nobel Prize-winning physicists in 1998.

  The project underwent rigorous review and approval by a committee composed of my advisor, Dr. Daniel Brinkman, and Dr. Dashiell Fryer. Throughout the research process, their expertise and guidance proved invaluable.

  During the presentation, which was open to the public, I discussed the transformation of fundamental cosmological equations into a dynamical system form, a cornerstone of the project. This transformation enabled analytical and numerical methods to explore the intricate dynamics of cosmological models. I emphasized investigating stability conditions within these models and elucidating their profound implications. The study of stability conditions played a pivotal role in selecting viable cosmological models for more extensive and detailed examinations. (San José State University, San José, California, May 2022)

Additional Education

- **MSRI Summer Graduate School** — Berkeley, CA
  Mathematics of Big Data: Sketching and (Multi-) Linear Algebra — *June 2021 – July 2021*
  Description: *Summer Graduate Course, Sketching-Based Approaches in Computational Linear and Multi-Linear Algebra*
  - **Attended** lectures introducing graduate students to sketching techniques for compressing matrices through the use of random linear maps.
  - **Studied** topics including matrix computations, tensor algebras, and applications and analysis of sketching techniques.

Honors and Awards

- GEM Fellowship Employer Select (2024): Selected by MIT Lincoln Laboratory
- Phi Kappa Phi Honor Society, June 2022
- Dean's List, Evergreen Valley College, Spring 2020
- Tau Beta Pi, Alpha Gamma Chapter (Engineering Honor Society), June 2012
- Second Runner-Up at IEEE Student Design Competition, October 2012

References

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